Linear Regression

Exercise 1

In this exercise, we will revisit the problem of performing linear regression to fit the measured concentration of CO_2 in parts per million in Hawaii over the time span between 1974 to 2020. The function which will be fitted in this exercise is of this structure.

$$g(t) = w_0 + w_1\phi_1(t) + w_2\phi_2(t) + ... + w_n\phi_n(t) \tag{1}$$

where $\phi_i(t)$ is a base function and w_i is a linear coefficient for $g(t)$. The base functions are known and the coefficients will be solved for using the least squares method but with different regularizers.

We will fit the function to the first half of the data and then see what happens when we extrapolate it into the future. Further, we will take a look at what can happen if the function is overfitted.

(a) Using the basis functions you chose earlier, solve for the coefficients with ridge regression and lasso regression for different regularization parameters λ.

(b) Create 100 different basis functions of your own choice and fit a new set of coefficients to the data in **data_fit** with the different regularization options. What happens now when you extrapolate the function into the future?

Helpful tips:

- Note, for the ridge regression, use the closed-form solution. For the Lasso regression, then either use CVX (`https://cvxr.com/cvx/doc/install.html#`) or Matlab's Problem-Based Optimization Workflow (`https://se.mathworks.com/help/optim/ug/optim.problemdef.optimizationproblem.html`).
- The original lasso problem is written as

$$\min_{w\in\mathbb{R}^M} \quad \frac{1}{2}\|\Phi(\boldsymbol{x})w - \boldsymbol{y}\|_2^2 + \lambda\|w\|_1.$$

  Since the L_1 norm is not differentiable at zero, we introduce auxiliary variables s_i so that $s_i \geq |w_i|$ for $i = 0, \dots, M-1$. This allows us to rewrite the L_1 term as $\sum_{i=0}^{M-1} s_i$ with linear constraints. The reformulated optimization problem becomes

$$\begin{aligned} \min_{w\in\mathbb{R}^M,\, s\in\mathbb{R}^M} \quad & \frac{1}{2}\|\Phi(\boldsymbol{x}) - \boldsymbol{y}\|_2^2 + \lambda\sum_{i=0}^{M-1} s_i, \\ \text{s.t.} \quad & -s_i \leq w_i \leq s_i, \quad i = 0, \dots, M-1, \\ & s_i \geq 0, \quad i = 1, \dots, M-1. \end{aligned}$$